ITSCA: SCIENTIFIC COMPUTING IN PYTHON

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SECTION A

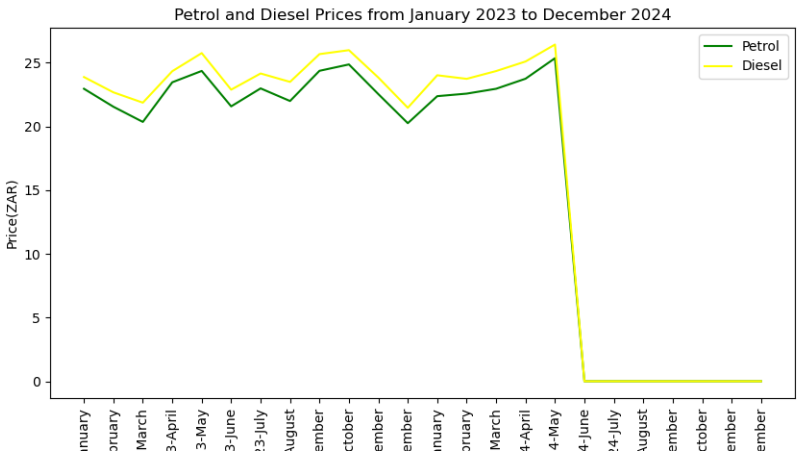
Question 1

Tracking petrol and diesel prices

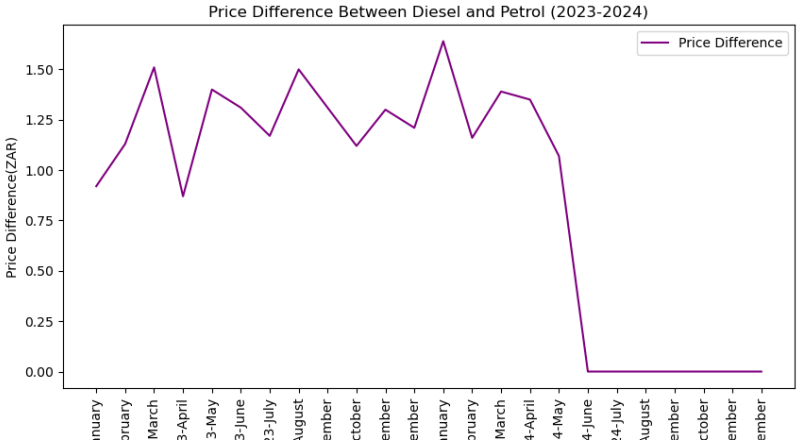
Additional Questions

What insights can be drawn from the price difference trends across the two years?

The analysis of fuel prices in South Africa from January 2023 to December 2024 reveals that there have been significant trends in petrol and diesel costs, this is crucial for budgeting. The dataset “petrolprices.csv” shows the price fluctuations, with data from May 2024 is incomplete, indication that there are potential missing or zero values.



In 2023, the diesel prices consistently exceed the petrol prices, with the differences ranging from R0.91 to R1.4. This indicates that diesel was more expensive than petrol. This particular trend suggests that a stable premium for diesel, which could impact transportation and industrial budgeting.



In 2024, the data shows zero values for prices after May, which can cause issues when trying to assess the full-year trend. This abrupt change can imply that there is missing data or a significant market event, such as a price freeze or data recording issues.

The consistent difference in 2023 indicates the need for households as well as businesses to account for this premium when forecasting fuel costs, while the incomplete 2024 data suggest caution in relying on current trends for future predictions.

While predicting the future is difficult, you can monitor trends and factors influencing fuel prices to prepare for potential increases.

(w3schools, 2020)

(Agarwal, 2018)

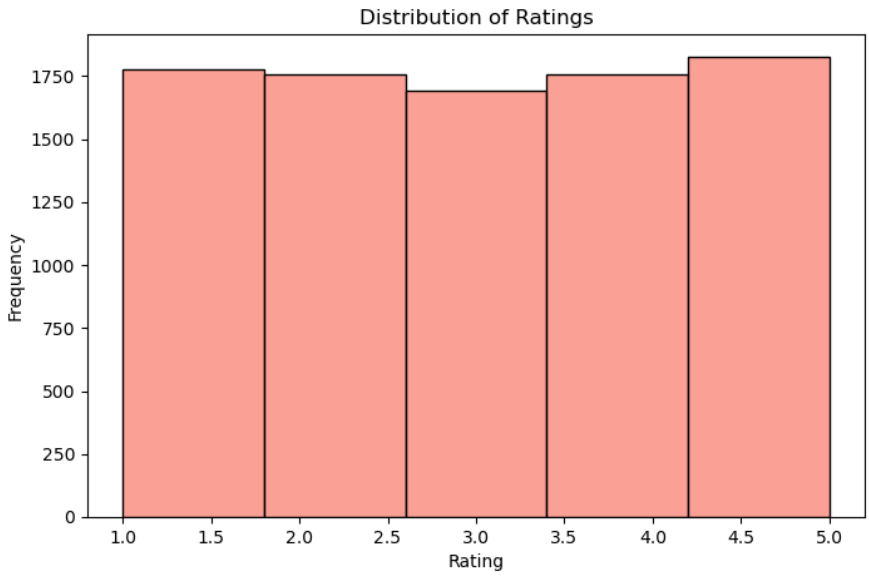
(Data Quest, 2024)

(w3schools, 2020)

Question 2

Sentiment Analysis of Series and Movie Descriptions

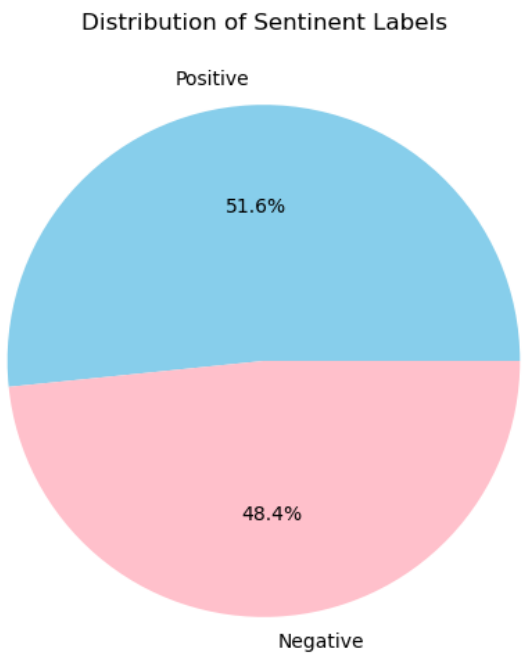
This analysis of “series_movies_descriptions.csv” aims to classify the emotional tone of the descriptions using computational techniques and explore their distributions



The histogram shows the Distribution of Ratings amongst the movies. The visualisation shows that there is an even spread across the 1-5 scale, with slight peaks at the lower (1.0 – 2.0) and higher (4.0 – 4.5) ratings, indicating a bimodal distribution, suggesting polarized ratings, while the middle range (e.g., 2.5-3.5) shows lower frequencies, indicating fewer neutral ratings.

Using TextBlob, sentiment polarity scores were computed which ranged from positive (1) to negative (-1). A new column “Polarity” was added to the data frame. The average polarity was approximately 0.12, indicating a slight positive lean in the descriptions.

There is also no significant bias toward any single rating category.



The pie chart Distribution of Sentiment Labels illustrates the proportion of positive and negative sentiments. The chart is divided into two sections: the larger section, representing 51.6% and labelled "Positive," is shown in light blue, while the smaller section, representing 48.4% and labelled "Negative," is shown in pink. This suggests a slight majority of positive sentiments, with the distribution being fairly balanced between positive and negative labels, indicating a nearly even split in the sentiment dataset.

(Gupta, 2023)

(Eduvos myLMS, 2025)

(GeeksForGeeks, 2025)

(w3Schools, 2020)

Additional Questions

Suggest one method to improve the accuracy of the sentiment classification, especially for longer or more nuanced descriptions

In order to improve the accuracy of the sentiment classification for a longer nuanced description is to ensemble methods that enhance predictive power by combining multiple classifiers, research shows up 20% improvement in precision. To classify sentiments effectively with machine learning, preprocess your text data by tokenizing, normalizing, removing stop-words, lemmatizing, and converting emojis and slang into text to eliminate noise. This approach captures the contextual nuances, and it can also capture sarcasm better than any rule-based tools, improving the classification for complex texts. Then, use supervised learning models like Support Vector Machines or Random Forests, which can achieve over 90% accuracy with tuned hyperparameters. Finally, apply cross-validation to ensure your model generalizes well across different data, creating a reliable sentiment classifier. (Crudu, 2025)

What limitations might exist in using rule-based tools like TextBlob for movie or series sentiment analysis?

Rule-based tools like TextBlob have limitations that include:

- the difficulty in detecting sarcasm,
- limited understanding of context in varying scenarios,
- inability to handle nuanced language,
- struggles with domain-specific jargon and slang

which can often be found in movie descriptions. They rely on lexical rules, which may misinterpret nuanced language, leading to less accurate sentiment scores.

(Crudu, 2025)

Question 3

Web Scraping for Job Market Data Analysis

The development of the web scraping python program is to retrieve job postings from the Career Junction website; it automates data extraction to support job market analysis through user interaction and data retrieval.

Users are prompted to input a job title, which forms the basis of a search URL. The script sends a request, parses the first page of search results, and extracts fields including Job Title, Recruiter Name, Salary, Position or Type, Job Location, and Date Posted. Extracted data is stored in a Pandas Data Frame and saved as a CSV file named after the inserted search term.

Scraping job data boosts recruitment through aggregation, faster updates, better matching, and competitive analysis.

(Coresignal, 2025)

(Amos, 2024)

(Zanini, n.d.)

Additional Questions

What challenges or limitations might arise when scraping dynamic websites such as CareerJunction?

Scraping dynamic websites comes with its challenges and limitations, with the Career Junction website specifically when loading content is requires JavaScript, which contains requests and BeautifulSoup cannot render. This may result in incomplete data unless a tool like Selenium can be used. There may be anti-scraping measures such as CAPTCHA blockers, Robots.txt and IP blocking., can block the requests. Frequents site updates or changes to the HTML structure can break the scraper, requiring regular maintenance and lastly there may be legal restrictions or terms of service that may prohibit scraping without consent. (Dilmegani, 2025)

How would you handle pagination to extract jobs beyond the first page?

Pagination is the process of dividing web content into discrete pages to make it more manageable and user-friendly. Analyse the URL structure of the pages. After identifying the URL pattern, you can loop through the pages by dynamically generating URLs. This involves iterating over a range of page numbers and constructing the URL for each page, allowing you to send requests and access the content of multiple pages sequentially. As it loops through each page, using Python libraries like BeautifulSoup or Scrapy to parse it and extract the required information, ensures the scraper handles each page consistently and captures all relevant data before moving to the next page.

(LinkedIn, 2025)

SECTION B

Question 4

Clustering Analysis for Customer Segmentation

The ”customer_segmentation.csv” dataset was loaded into a Pandas DataFrame. Some rows within the dataset had null values, which were removed, and only ”Income” and ”Spending Score” columns were retained for analysis.

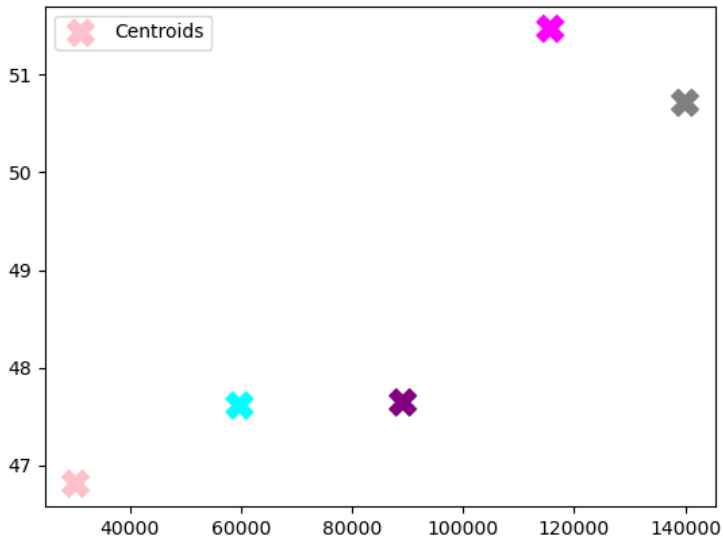
The elbow method, a technique used to determine the optimal number of clusters within a dataset, was applied using the KMeans algorithm from the scikit-learn library to analyse the customer segmentation data. It resulted in 5 optimal clusters based on the Within-Cluster Sum of Squares (WCSS). A detailed scatter plot illustrates the data distribution and relationship between the “Income” and ‘Spending Score” features, which highlights patterns and potential groupings among the customers. This was followed by the creation of a comprehensive cluster plot that not only illustrated the five distinct clusters using different colours for each group but also prominently marked the centroids of these clusters with a distinct marker. These centroids, representing the central point of each cluster, provided a clear focal point for understanding the average characteristics of the customers within each segment, thereby enhancing the interpretability and utility of the clustering results for subsequent marketing strategies.

(GeeksForGeeks, 2025)

(Banoula, 2025)

Additional Questions

Write a short summary (3–5 lines) explaining the number of clusters formed and the apparent characteristics of each cluster



Analysing the clustering, it has formed 5 distinct customer segments based on Income and Spending Score. Cluster 1 represents high-income, low-spending customers; Cluster 2 represents low-income and high-spending customers; Cluster 3 represents average-income, average-spending customers; Cluster 4 represents low-income and low-spending customers and Cluster 5 represents high-income and high-spending customers.

Discuss how this clustering output might assist the retailer in tailoring advertisements for different customer groups.

The result of the clusters will enable retailers to tailor advertisements by targeting Cluster 5 with premium product promotions, whereas Cluster 2 will require discounts to enhance their loyalty, for Cluster 1 tailored advertisements include investments and savings offers. Cluster 4 customers can be approached with budget-friendly campaigns, while Cluster 3 can receive balanced value propositions, optimizing as effectiveness and personalization across segments.

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